Practice Sheet 02

1. Limit proofs by ε–δ

(a) $\lim\limits_{(x,y)\to(1,2)} (5x^3 - x^2y^2)$

Proof. The function is a polynomial, hence continuous. Substituting gives $5 \cdot 1^3 - 1^2 \cdot 2^2 = 5 - 4 = 1$. Let $\varepsilon > 0$. Choose $\delta = \min\{1, \varepsilon/30\}$. For $\|(x,y) - (1,2)\| < \delta$ we have $|x-1| < \delta$ and $|y-2| < \delta$. Using $|x^3 - 1| \le 7|x-1|$ and $|x^2y^2 - 4| \le 20\|(x,y) - (1,2)\|$ for $\delta \le 1$, one obtains

$$|5x^3 - x^2y^2 - 1| \le 30\|(x,y) - (1,2)\| < \varepsilon.$$

Thus the limit is 1. □

(b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2+y^2}}$

Proof. We claim the limit is 0. For any $(x,y) \neq (0,0)$,

$$\left|\frac{xy}{\sqrt{x^2+y^2}}\right| \le \frac{|x||y|}{\sqrt{x^2+y^2}} \le \frac{(\sqrt{x^2+y^2})^2}{\sqrt{x^2+y^2}} = \sqrt{x^2+y^2}.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$. Then $\sqrt{x^2+y^2} < \delta$ implies the expression is $< \varepsilon$. Hence the limit is 0. □

(c) $\lim\limits_{(x,y)\to(2,1)} \dfrac{4-xy}{x^2+3y^2}$

Proof. The function is continuous at $(2,1)$ because denominator $x^2 + 3y^2 > 0$. Direct substitution gives $\frac{4-2\cdot 1}{4+3} = \frac{2}{7}$. For an ε–δ proof, choose $\delta = \min\{1, \frac{49\varepsilon}{40}\}$. For $\|(x,y) - (2,1)\| < \delta$ we have $|x-2|, |y-1| < \delta$ and $x^2 + 3y^2 \ge 4$. Then

$$\left|\frac{4-xy}{x^2+3y^2} - \frac{2}{7}\right| = \frac{|7(4-xy) - 2(x^2+3y^2)|}{7(x^2+3y^2)} \le \frac{\text{linear in } \delta}{28} < \varepsilon.$$

Thus the limit is 2/7. □

(d) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}}$

Proof. Let $r = \sqrt{x^2+y^2}$. The function becomes $\frac{r^2}{\sqrt{r^2+1}}$. As $r \to 0$, this tends to 0. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $r < \delta$ implies

$$\left|\frac{x^2+y^2}{\sqrt{x^2+y^2+1}}\right| \le r^2 < \delta^2 \le \varepsilon \quad \text{if } \delta \le \sqrt{\varepsilon}.$$

A simpler choice: $\delta = \min\{1, \varepsilon\}$ works because $r^2 \le r < \varepsilon$. Hence the limit is 0. □

(e) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}$

Proof. For $(x,y)\neq(0,0)$,

$$\left|\frac{3x^2y}{x^2+y^2}\right| \le 3|y| \le 3\sqrt{x^2+y^2}.$$

Given $\varepsilon>0$, take $\delta=\varepsilon/3$. Then $\sqrt{x^2+y^2}<\delta$ gives the expression $<3\delta=\varepsilon$. Thus the limit is 0. □

(f) $\displaystyle\lim_{(x,y)\to(2,2)} \frac{x^3-y^3}{x^2-y^2}$

Proof. For $x\neq y$,

$$\frac{x^3-y^3}{x^2-y^2} = \frac{(x-y)(x^2+xy+y^2)}{(x-y)(x+y)} = \frac{x^2+xy+y^2}{x+y}.$$

The function is continuous at $(2,2)$ (since denominator $x+y\neq0$). Substituting gives $\frac{4+4+4}{4}=3$. For an ε–δ proof, choose $\delta=\min\{1,\varepsilon/9\}$. Using the bounds $|x-2|,|y-2|<\delta$ one can show

$$\left|\frac{x^2+xy+y^2}{x+y}-3\right| \le 9\delta<\varepsilon.$$

Hence the limit is 3. □

(g) $\displaystyle\lim_{(x,y)\to(1,1)} \frac{2(x-y^2)}{x^2-y^4}$

Proof. For $x\neq y^2$, factor:

$$\frac{2(x-y^2)}{x^2-y^4} = \frac{2(x-y^2)}{(x-y^2)(x+y^2)} = \frac{2}{x+y^2}.$$

The function $\frac{2}{x+y^2}$ is continuous at $(1,1)$ because $x+y^2=2\neq0$. Substituting gives $\frac{2}{1+1}=1$. For the ε–δ argument, choose $\delta=\min\{1/4,\varepsilon/4\}$. Then for $\|(x,y)-(1,1)\|<\delta$ we have $|x-1|,|y-1|<\delta$, so

$$|x+y^2-2| \le |x-1|+|y-1||y+1| < \delta+\delta\cdot3 = 4\delta,$$

and $x+y^2>1$. Hence

$$\left|\frac{2}{x+y^2}-1\right| = \frac{|2-(x+y^2)|}{x+y^2} < \frac{4\delta}{1} = 4\delta \le \varepsilon.$$

Thus the limit is 1. □

2. $\displaystyle\lim_{(x,y)\to(0,0)} \frac{5xy^2}{x^2+y^2}=0$

Proof. For $(x,y)\neq(0,0)$,

$$\left|\frac{5xy^2}{x^2+y^2}\right| \le 5|y|\cdot\frac{|x|}{x^2+y^2}\cdot|y|.$$

Better: $\left|\frac{5xy^2}{x^2+y^2}\right| \le 5|y|$ because $|x|\le\sqrt{x^2+y^2}$ and $\frac{|x|}{\sqrt{x^2+y^2}}\le1$. Actually,

$$\frac{|x|y^2}{x^2+y^2} \le |y|.$$

Thus $\left|\frac{5xy^2}{x^2+y^2}\right| \le 5|y| \le 5\sqrt{x^2+y^2}$. Given $\varepsilon>0$, choose $\delta=\varepsilon/5$. Then $\sqrt{x^2+y^2}<\delta$ implies the expression $<5\delta=\varepsilon$. Hence the limit is 0. □

3. $f(x, y) = \dfrac{xy^2}{x^2 + y^4}$

Proof. We investigate the limit as $(x, y) \to (0, 0)$. Along the line $x = 0$ we get $f(0, y) = 0 \to 0$. Along the parabola $x = y^2$ we get

$$f(y^2, y) = \frac{y^2 \cdot y^2}{(y^2)^2 + y^4} = \frac{y^4}{y^4 + y^4} = \frac{1}{2}.$$

Since we obtain different limits along different paths (0 and 1/2), the limit does not exist. Therefore $\lim\limits_{(x,y)\to(0,0)} f(x, y)$ does not exist. $\square$

4. Continuity of $f(x, y) = \begin{cases} \frac{\cos y \sin x}{x}, & x \neq 0 \\ \cos y, & x = 0 \end{cases}$ at $(0, 0)$

Proof. For $x \neq 0$, $f(x, y) = \cos y \cdot \frac{\sin x}{x}$. As $x \to 0$, $\frac{\sin x}{x} \to 1$. Hence for any fixed y, $\lim_{x\to 0} f(x, y) = \cos y$. For $(x, y) \to (0, 0)$, we examine

$$|f(x, y) - f(0, 0)| = \left|\cos y \cdot \frac{\sin x}{x} - \cos 0\right| = \left|\cos y \cdot \frac{\sin x}{x} - 1\right|.$$

Write

$$\cos y \cdot \frac{\sin x}{x} - 1 = \cos y \left(\frac{\sin x}{x} - 1\right) + (\cos y - 1).$$

Given $\varepsilon > 0$, choose $\delta > 0$ such that $|x| < \delta$ implies $|\frac{\sin x}{x} - 1| < \varepsilon/2$ (possible by the single-variable limit), and also such that $|y| < \delta$ implies $|\cos y - 1| < \varepsilon/2$ (by continuity of cosine). Then for $\|(x, y)\| < \delta$,

$$|f(x, y) - 1| \leq |\cos y| \cdot \left|\frac{\sin x}{x} - 1\right| + |\cos y - 1| < 1 \cdot \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $\lim_{(x,y)\to(0,0)} f(x, y) = 1$. But $f(0, 0) = \cos 0 = 1$, so f is continuous at $(0, 0)$. No redefinition is needed. $\square$

Additional Problems (Continuity & Sequences)

1. $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is not continuous at $(0, 0)$

Proof. Consider the sequence $(x_k, y_k) = (\frac{1}{k}, \frac{1}{k})$. Then $f(x_k, y_k) = \frac{1/k^2}{2/k^2} = \frac{1}{2} \to \frac{1}{2} \neq 0 = f(0, 0)$. Hence f is not continuous at $(0, 0)$. $\square$

2. $f(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

Proof. Take any fixed line through $(0, 0)$: either $x = 0$ or $y = mx$ (or $x = my$). - If $x = 0$, then $f(0, y) = 0 \to 0$. - If $y = mx$, then $f(x, mx) = \frac{x \cdot m^2 x^2}{x^2+m^4x^4} = \frac{m^2x^3}{x^2+m^4x^4} = \frac{m^2x}{1+m^4x^2} \to 0$ as $x \to 0$. Thus along any line through the origin, the limit is 0. However, the function is not continuous because along the parabola $x = y^2$ we get $f(y^2, y) = \frac{y^2 \cdot y^2}{y^4+y^4} = \frac{1}{2} \neq 0$. Hence f is not continuous at $(0, 0)$. $\square$

3. $g(x,y) = \begin{cases} x\sin\frac{1}{y}, & y \neq 0 \\ 0, & y = 0 \end{cases}$

Proof. We examine continuity at an arbitrary point (x_0, y_0). - If $y_0 \neq 0$, then near (x_0, y_0) the function is a product of continuous functions (since $y \mapsto 1/y$ is continuous away from 0), so g is continuous at (x_0, y_0). - If $y_0 = 0$, then $g(x, 0) = 0$. For continuity at $(x_0, 0)$, we need $\lim_{(x,y)\to(x_0,0)} g(x, y) = 0$. For any (x, y) with $y \neq 0$, $|g(x, y)| = |x \sin \frac{1}{y}| \leq |x|$. As $(x, y) \to (x_0, 0)$, we can make $|x|$ arbitrarily close to $|x_0|$, but not necessarily small. However, we can estimate: $|g(x, y) - 0| = |x \sin \frac{1}{y}| \leq |x|$. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $\|(x, y) - (x_0, 0)\| < \delta$ implies $|x - x_0| < \delta$ and $|y| < \delta$. But $|x|$ could be as large as $|x_0| + \delta$, which is not necessarily small. This naive bound fails when $x_0 \neq 0$. Actually, the function is **not** continuous at points $(x_0, 0)$ with $x_0 \neq 0$. Consider a sequence $(x_n, y_n) = (x_0, \frac{1}{n})$. Then $g(x_n, y_n) = x_0 \sin n$, which does not converge (oscillates). Hence g is not continuous at such points. At $(0, 0)$ we have $g(0, 0) = 0$ and $|g(x, y)| \leq |x| \leq \sqrt{x^2 + y^2}$, so g is continuous at $(0, 0)$. Thus g is continuous at all points with $y \neq 0$, and at $(0, 0)$, but not at points $(x_0, 0)$ with $x_0 \neq 0$. □

Theorem on Iterated Limits

Proof of the theorem. Suppose $\lim_{(x,y)\to(a,b)} f(x, y) = L$ and the one-dimensional limits $\lim_{x\to a} f(x, y) =: \phi(y)$ and $\lim_{y\to b} f(x, y) =: \psi(x)$ exist for each y (resp. x) in a neighbourhood. We show $\lim_{y\to b} \phi(y) = L$. Given $\varepsilon > 0$, choose $\delta > 0$ such that $0 < \|(x, y)-(a, b)\| < \delta$ implies $|f(x, y)-L| < \varepsilon/2$. For a fixed y with $0 < |y-b| < \delta$, take $x \to a$ in the inequality $|f(x, y)-L| < \varepsilon/2$ (valid for all x with $0 < |x-a| < \sqrt{\delta^2 - |y - b|^2}$). By the definition of $\phi(y) = \lim_{x\to a} f(x, y)$, we obtain $|\phi(y) - L| \leq \varepsilon/2 < \varepsilon$. Thus $\lim_{y\to b} \phi(y) = L$. The other equality follows similarly. □

(b) Example with different iterated limits

$$f(x,y) = \begin{cases} 1, & x \neq 0,\ y = 0, \\ -1, & x = 0,\ y \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\lim_{x\to 0}\big(\lim_{y\to 0} f(x, y)\big) = 1$ but $\lim_{y\to 0}\big(\lim_{x\to 0} f(x, y)\big) = -1$.

(c) Example where iterated limits are 0 but double limit does not exist

$$f(x,y) = \begin{cases} 1, & y = x^2,\ x \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

For each fixed $x \neq 0$, $\lim_{y\to 0} f(x, y) = 0$; for each fixed y, $\lim_{x\to 0} f(x, y) = 0$. Hence both iterated limits are 0. But along the parabola $y = x^2$, $f(x, x^2) = 1$, so the double limit does not exist.